

3.9 Organic Synthesis

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.9 Organic Synthesis
Difficulty	Hard

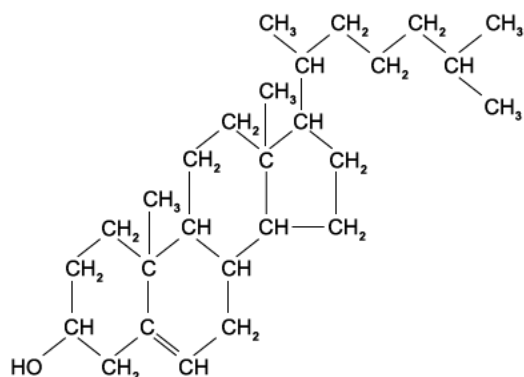
Time allowed: 20

Score: /10

Percentage: /100

Question 1

The diagram shows the structure of the naturally-occurring molecule cholesterol.



Two separate test tubes of cholesterol are used in an experiment. One test tube of cholesterol is treated with cold, dilute acidified KMnO_4 , and the other is treated with hot, concentrated acidified KMnO_4 .

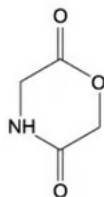
Which row correctly describes the change in the **number** of chiral carbon atoms in the molecule during each reaction and the structure of the product in each test tube?

	Cold, dilute acidified KMnO_4		Hot, concentrated acidified KMnO_4	
	Change in the number of chiral carbons	Number of hydroxy groups in product	Change in the number of chiral carbons	Number of 6-membered rings remaining in product
A	+1	1	0	2
B	+1	1	-1	3
C	+2	3	-1	2
D	+2	3	0	3

[1 mark]

Question 2

The cyclic compound G is heated with dilute hydrochloric acid.



Compound G

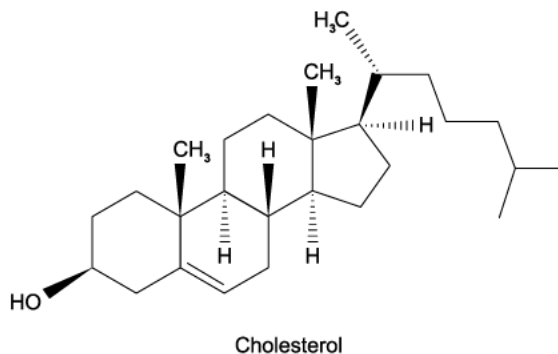
What are the products of the reaction?

- A. $\text{HOCH}_2\text{CO}_2\text{H}$ and $\text{H}_2\text{NCH}_2\text{CO}_2\text{H}$
- B. $\text{H}_2\text{NCOCH}_2\text{OH}$ and HOCH_2CHO
- C. $\text{HOCH}_2\text{CONH}_3^+$ and HOCH_2CHO
- D. $\text{HO}_2\text{CCH}_2\text{NH}_3^+$ and $\text{HO}_2\text{CCH}_2\text{OH}$

[1 mark]

Question 3

Cholesterol found in the cell membrane of animal cells has the following structure.



Which statements are correct?

- 1 Cholesterol forms an aldehyde after oxidation by acidified sodium dichromate (VI) to
- 2 Cholesterol reacts with bromine to form a compound which has two new chiral centres
- 3 Cholesterol reacts with a mixture of concentrated sulphuric acid and ethanoic acid

A. 1, 2 and 3

B. 1 and 2

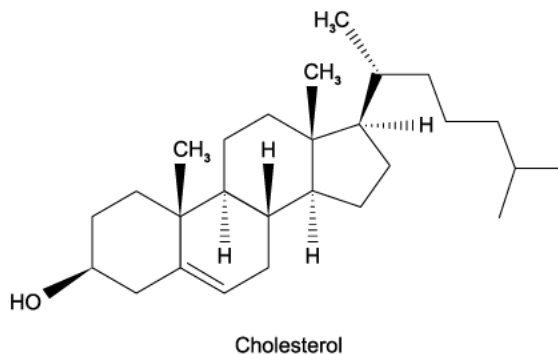
C. 2 and 3

D. 1 only

[1 mark]

Question 4

The structure of the naturally-occurring molecule cholesterol is shown below.



Student 1 says that cholesterol displays *cis-trans* isomerism at the C = C double bond.

Student 2 says all carbon atoms within the rings of cholesterol all lie in the same plane.

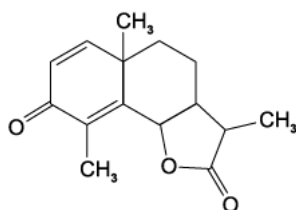
Which of the students are correct?

- A. Both 1 and 2
- B. 1 only
- C. 2 only
- D. Neither 1 nor 2

[1 mark]

Question 5

Santonin is a drug that has been used to treat a worm infection.



Santonin

When santonin is first treated with warm dilute H₂SO₄, product X is formed. If X is treated with cold acidified KMnO₄, the final product Y is obtained.

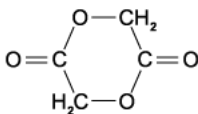
How many atoms of hydrogen in each molecule of product Y could be displaced in a reaction with sodium metal?

- A. 2
- B. 4
- C. 5
- D. 6

[1 mark]

Question 6

Compound M is used in leather processing.



Compound M

Compound M is synthesised in the presence of concentrated sulfuric acid and heat.

Which other starting material(s) are required to synthesise compound M?

- A. CH₃COOH only
- B. HOCH₂COOH only
- C. CH₃COOCH₂COOH only
- D. CH₃COOH mixed with HOCH₂COOH

[1 mark]

Question 7

Mixture **F** is produced upon refluxing compound **G** with aqueous sodium hydroxide.

Upon distillation with acidified potassium dichromate (VI) propanone is produced. Mixing **J** with aqueous silver nitrate and dilute nitric acid gives a cream precipitate.

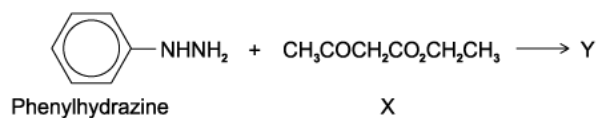
What could compound **G** be?

- A. $\text{CH}_3\text{CHICH}_3$
- B. $\text{CH}_3\text{CHBrCH}_3$
- C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{I}$
- D. $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$

[1 mark]

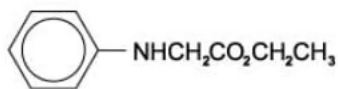
Question 8

Antipyrine is a drug used to reduce fever. The initial step of its synthesis is the reaction of phenylhydrazine with compound **X**.



What is the product **Y**?

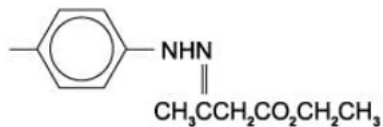
A.



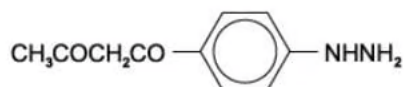
B.



C.



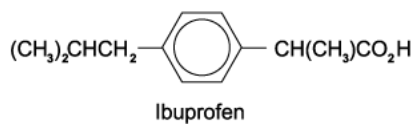
D.



[1 mark]

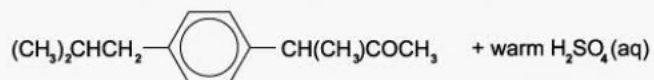
Question 9

Ibuprofen is an anti-inflammatory drug, the structure of which is shown below.

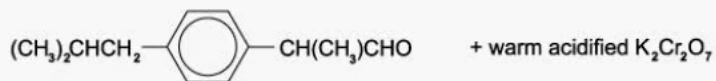


Which reaction leads to the formation of ibuprofen?

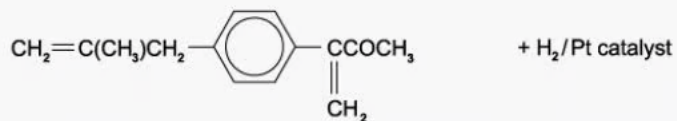
A.



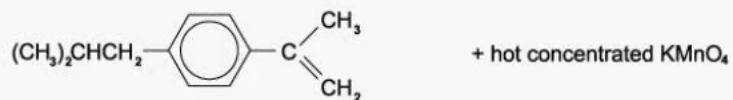
B.



C.



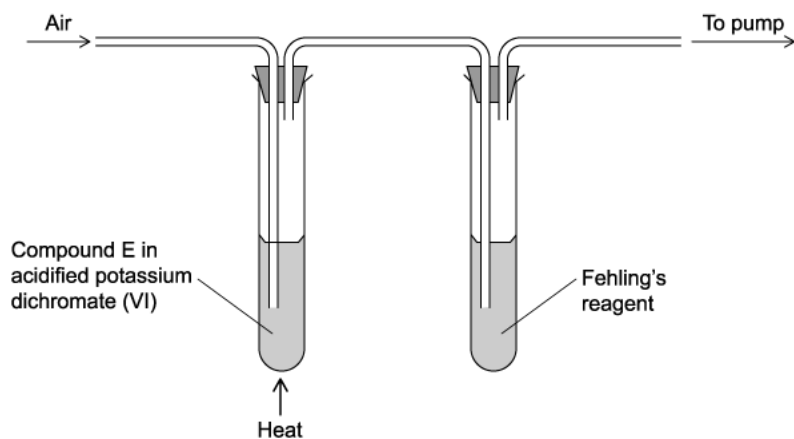
D.



[1 mark]

Question 10

When the apparatus below was used with compound **E**, a brick-red precipitate formed in the right-hand tube.



Which compound could be **E**?

- 1 CH_3OH
- 2 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- 3 $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

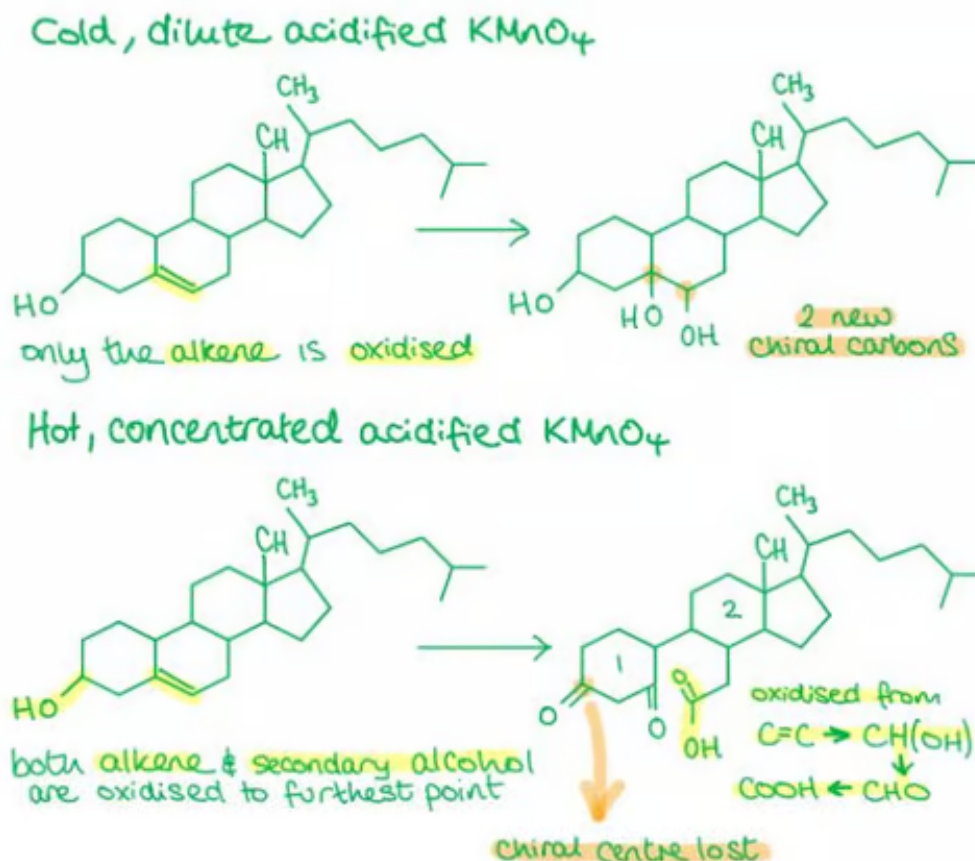
[1 mark]

Model Answers: Hard

1

The correct answer is **C** because:

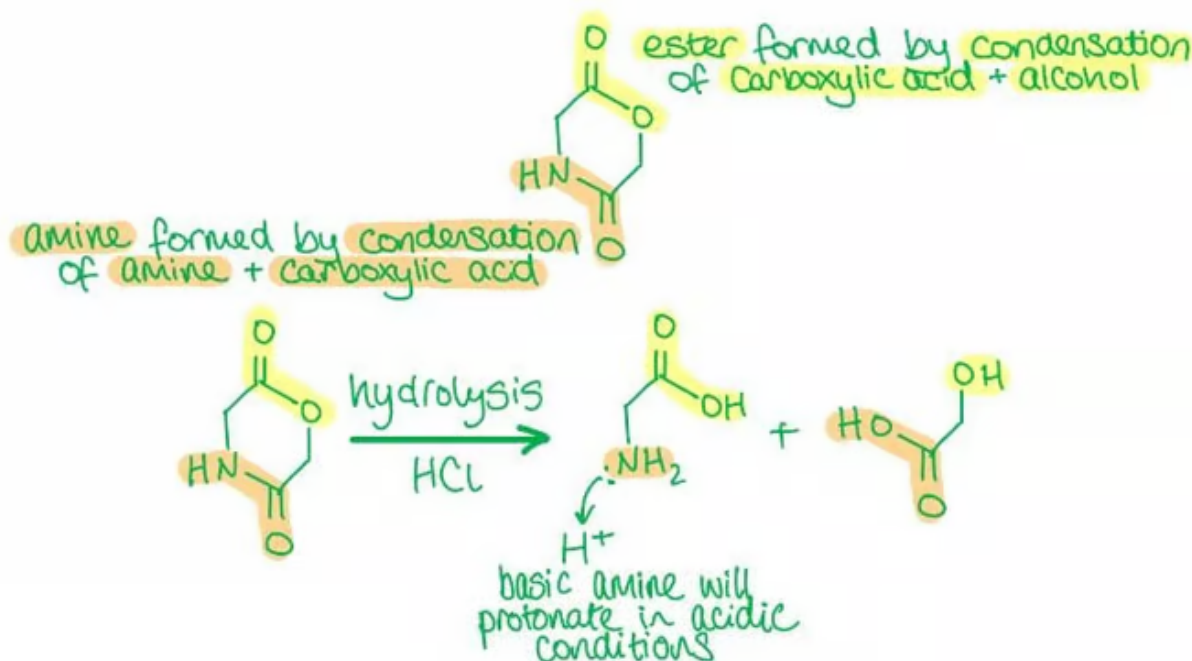
- **Cold, dilute MnO_4^-** is a **mild** oxidising reagent which oxidises alkenes to a diol.
- The **alkene** in cholesterol will become a **diol** and the two carbon atoms will now be bonded to four different groups causing an increase of **two chiral centres**.
- Cold, dilute MnO_4^- will not oxidise the $-\text{OH}$ group that is already present in cholesterol.
- Therefore, the **total number of hydroxy groups** in the product is **three**.
- **Hot, concentrated MnO_4^-** is a **strong** oxidising reagent which causes the $\text{C}=\text{C}$ bond to completely break to form a ketone or aldehyde.
- The $\text{C}=\text{C}$ in cholesterol will be oxidised to form a ketone and an aldehyde which is further oxidised to a carboxylic acid.
- The secondary alcohol group which is already present in cholesterol will get oxidised to a ketone.
- This will **reduce** the number of **chiral centres by 1** and only **two 6-membered rings** remain.



2

The correct answer is **D** because:

- Compound G contains an **amide** and **ester** group which can undergo **hydrolysis** when heated with diluted hydrochloric acid.
- Hydrolysis is breaking a compound down using **water**.
- Hydrolysis of the amide will form an **amine** and **carboxylic acid**.
 - Under acidic conditions, the basic amine will be protonated.
- Hydrolysis of an **ester** will form an **alcohol** and **carboxylic acid**.



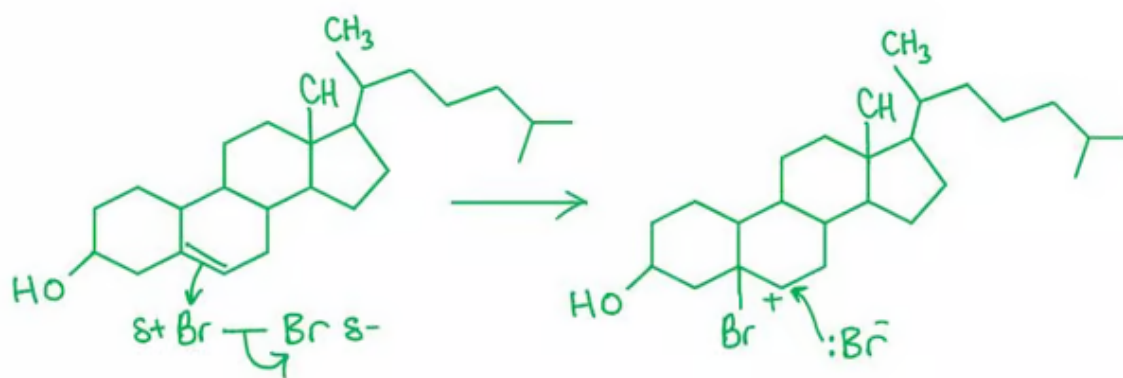
A is incorrect as there is **no alcohol group** which can form an ester with a carboxylic acid. The primary amine is protonated under acidic conditions.

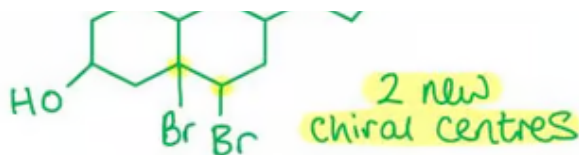
B & C are incorrect as the presence of the **ether** (C-O-C) will result in an **additional oxygen** atom in the ring. The primary amine is protonated under acidic conditions.

3

The correct answer is **C** because:

- **Alkenes** will undergo an **electrophilic addition** reaction when shaken with orange/yellow bromine water (Br_2) to turn the solution **colourless**.
- Since cholesterol contains an alkene group, it will undergo an addition reaction with Br_2 and due to the loss of the double bond, the two carbon atoms have now **four different groups** resulting in two new chiral centres in cholesterol.
- Cholesterol has an alcohol group ($-\text{OH}$) which will undergo **acid/base** reactions with concentrated sulphuric and ethanoic acid.





A, B & D are incorrect as acidified dichromate oxidises primary alcohols, secondary alcohols and aldehydes to carboxylic acids, ketones and carboxylic acids respectively. Acidified dichromate is not strong enough of an oxidising agent to oxidise alkenes.

4

The correct answer is **D** because:

- The double bond is part of a **six-membered ring** which forces the carbon atoms to be only in **one** possible configuration.
- The double bond can therefore **not** display **cis or trans isomers**.
- Only carbons in a triple or double bond lie in the same plane.
- The carbon atoms that make up the rings in cholesterol are bonded to each other by **sigma bonds** around which **rotation is possible**.
- Therefore, not all carbon atoms in cholesterol lie in the same plane.

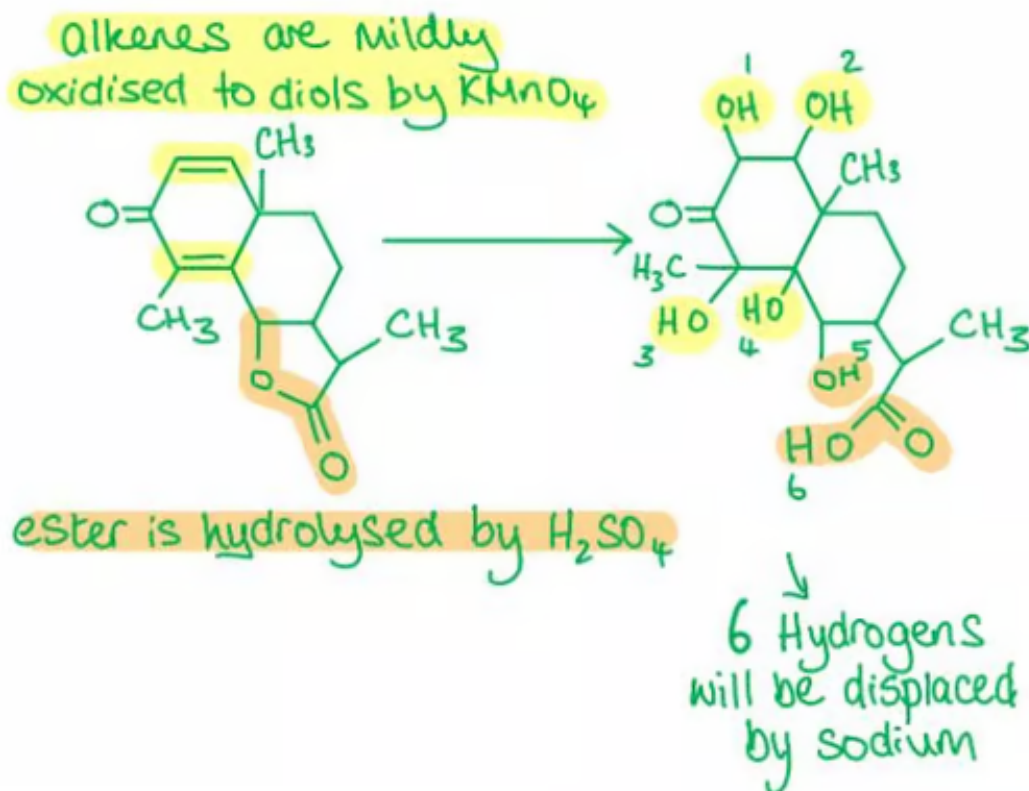
5

The correct answer is **D** because:

- When an **ester** is heated under **reflux** with a dilute acid such as H_2SO_4 , the ester will react with the water present in a **hydrolysis reaction** to form a carboxylic acid and an alcohol.
- **Alkenes** will undergo **mild oxidation** by cold, dilute acidified MnO_4^- to form a **diol** (2 alcohol groups).
- Since there are two alkene groups present in Santonin **two diols** will be formed from the mild oxidation of these alkenes.
 - Cold, dilute MnO_4^- is not strong enough of an **oxidising** reagent to oxidise the alcohol, produced from the hydrolysis of the ester, to a carboxylic acid.
- This gives a total of six (four hydrogens from the diols, and one hydrogen each from

the carboxylic acid and alcohol formed by the hydrolysis of the ester) hydrogen atoms which can be displaced in a reaction with sodium metal.

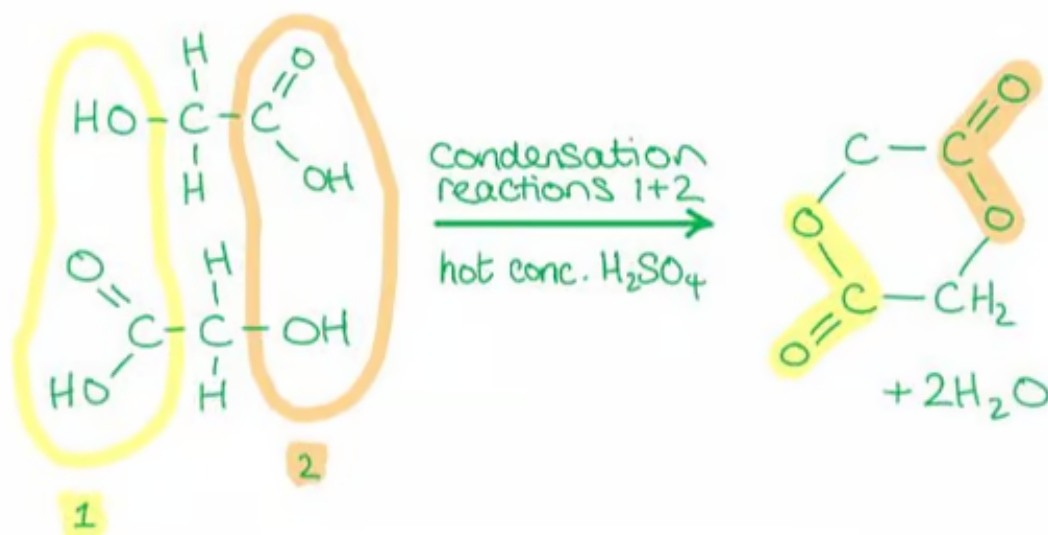
- **Alcohols** and **carboxylic acids** react with **reactive metals** such as Na to form a **salt and water**.



6

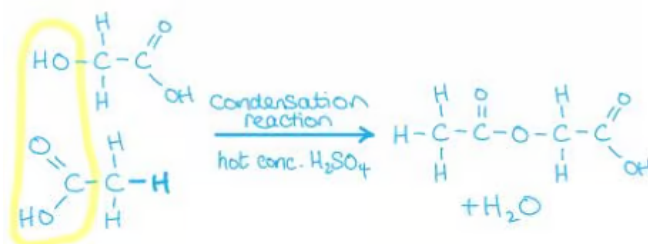
The correct answer is **B** because:

- Two molecules of HOCH_2COOH will react with each other to form **two ester groups**.
- An ester group is formed from the **condensation** reaction of an **alcohol** and **carboxylic acid** in which a water molecule is eliminated.



A & C are incorrect as there is no alcohol group which can react with a carboxylic acid of another molecule to form an ester.

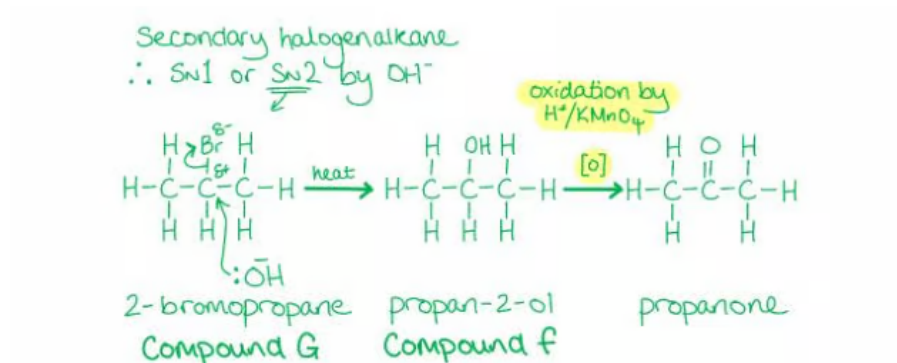
D is incorrect as this will not result in the formation of two ester groups.



7

The correct answer is **B** because:

- Since, propanone is produced upon distillation of mixture **F** with acidified potassium dichromate, **F** must have been a **secondary alcohol**.
- **Propan-2-ol (F)** forms when compound **G** had reacted with aqueous NaOH.
- Nucleophilic substitution occurs in which the OH⁻ substitutes a halogen.
- Compound **G** is therefore a **halogenoalkane** which react with silver nitrates to form a coloured precipitate of **silver halides**.
- **Silver bromide** is a **creamy precipitate**, therefore the unknown halogen is **bromine**.

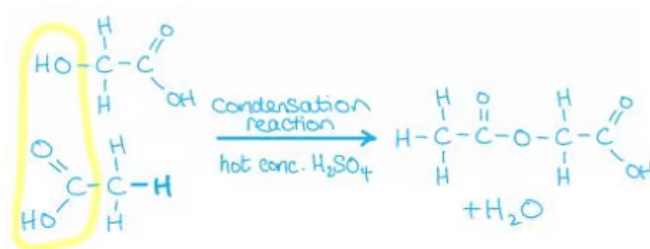


A is incorrect as though this is a secondary halogenoalkane, iodine doesn't give a



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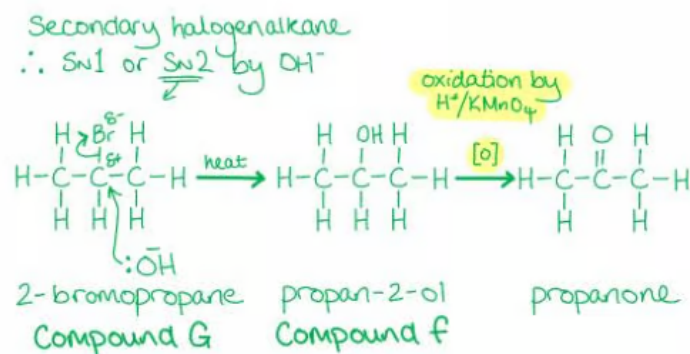
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A is incorrect as though this is a secondary halogenoalkane, iodine doesn't give a creamy precipitate when reacted with silver nitrate.

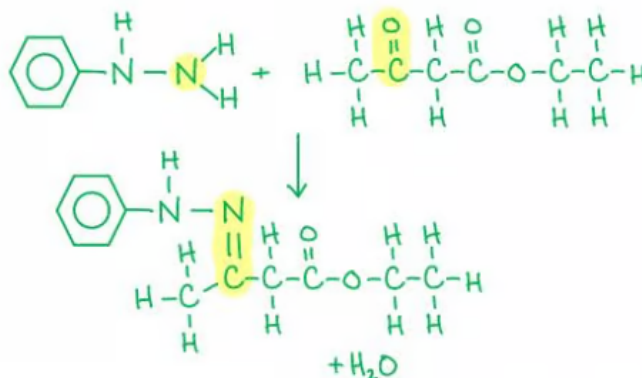
C is incorrect as this is a **primary halogenoalkane** which will form a **primary alcohol** when reacted with $NaOH$. The **primary alcohol** will get oxidised to **propanoic acid** and not propanone. Iodine doesn't give a creamy precipitate with silver nitrate.

D is incorrect as this is a **primary halogenoalkane** which will form a **primary alcohol** when reacted with $NaOH$. The **primary alcohol** will get oxidised to **propanoic acid** and not propanone.

8

The correct answer is **C** because:

- Phenylhydrazine contains an **amine group** will carry out a **condensation** reaction the ketone group of compound **X**.
 - The oxygen is more **electronegative** and has a greater pull on the electrons in the $C=O$ bond leaving the carbon slightly positive and the oxygen slightly negatively charged.
 - The nitrogen's **lone pair** adds on to the slightly positively charged carbonyl carbon.
 - A water molecule is eliminated.
- This will result in the formation of compound **C**.



A is incorrect as the amine group in phenylhydrazine and the CH_3CO group in compound **X** have been lost.

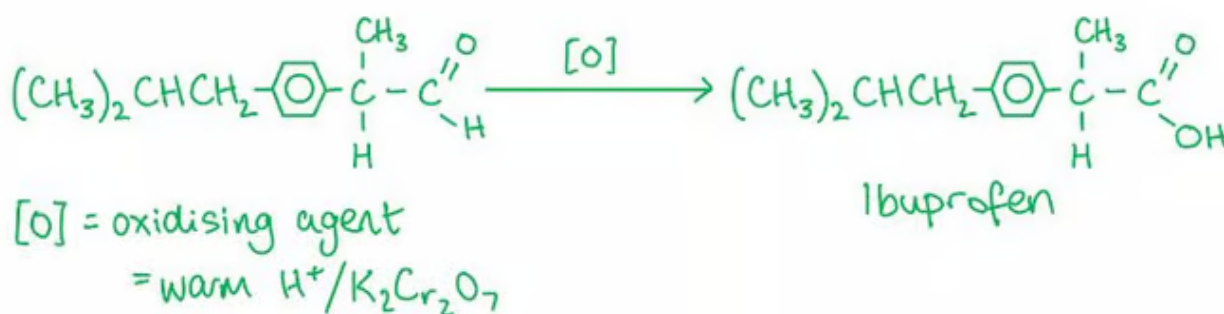
B is incorrect as the amine group in phenylhydrazine has been lost.

D is incorrect as the amine and ketone functional groups will react with each other.

9

The correct answer is **B** because:

- Hot acidified potassium dichromate (VI) **oxidises** primary alcohols, secondary alcohols and aldehydes to carboxylic acids, ketones and carboxylic acids respectively.
- The aldehyde in compound **B** will be oxidised to a **carboxylic acid**.



A & **D** are incorrect as alkenes undergo strong oxidation with **hot, concentrated** MnO_4^- in which the $\text{C}=\text{C}$ fully breaks. The compound is oxidised to a **ketone** and CO_2 .

C is incorrect as the two alkenes in the compound will react with H_2 (hydrogenation reaction) in the presence of a Pt catalyst.

10

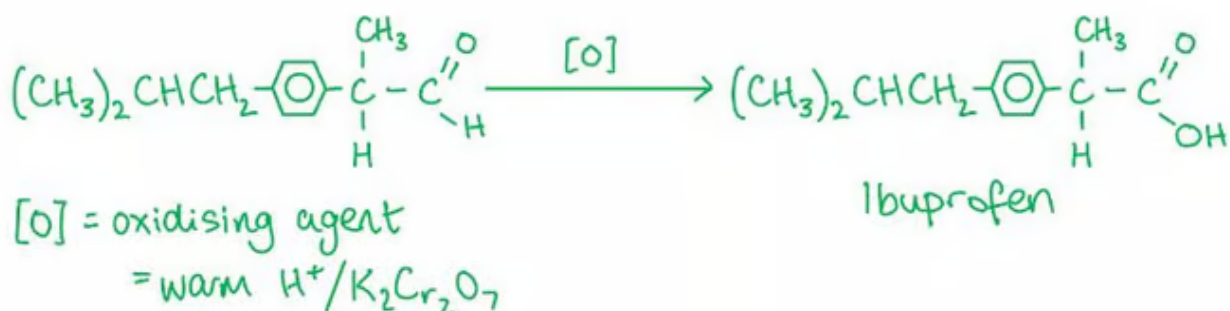
The correct answer is **B** because:

- The brick-red precipitate is copper (I) oxide which forms when an **aldehyde** is oxidised to a **carboxylic acid** and the Cu^{2+} ions are reduced to Cu^+ ions in Fehling's solution.
- Compound E must therefore have been an **aldehyde** which will get oxidised by the

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- Compound **E** must therefore have been an **aldehyde** which will get oxidised by the Fehling's reagent to a carboxylic acid.
- Compound **E** must therefore have been a **primary alcohol** which is oxidised to an **aldehyde** by acidified potassium dichromate (VI) in the left-hand tube.
- Both compound **1** and **2** are primary alcohols which can be oxidised to aldehydes

which will give brick-red precipitates in Fehling's solution.

A & C are incorrect as compound **3** is a secondary alcohol which will get oxidised to a **ketone** in the left-hand tube. Ketones don't form a brick-red precipitate in Fehling's solution.

D is incorrect as compound **E** could also be $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$.

